Career Launcher MBA

QA Arithmetic – 09

Proportionality; Average Speed

Question 1

A car travels from A to B in 46 min. If the distance between A and B is 52.8 km, what is the (approximate) average speed of the car?

(1) 1.14 km/hr

(2) 40.5 km/hr

(3) 68.86 km/hr

(4) 84.2 km/hr

Question 2

A person covers 180 km, 240 km and 300 km by car, train and bus respectively. Find the average speed of the whole journey if the speed of car, train and bus is 60 km/hr, 80 km/hr and 50 km/hr respectively.

(1) $63\frac{1}{3}\text{km/hr}$

(2) 60 km/hr

(3) $62\frac{1}{3}\text{km/hr}$

(4) 62 km/hr

Question 3

Pravin goes from Mumbai to Pune at a speed of 40 kmph and comes back at a speed of 60 kmph, taking exactly the same path. Find his average speed for the entire journey.

(1) 50 kmph

(2) 48 kmph

(3) 52 kmph

(4) 54 kmph

Question 4

Mandar travelled first 1/3rd distance of his journey at a speed of 40 kmph, 1/3rd of the remaining distance at 50 kmph and the rest at 90 km/hr. Find his average speed.

(1) 60 kmph

(2) 54 kmph

(3) 58 kmph

(4) None of these

Question 5

A car travels at 50 km/hr for the first half time of the journey, at 25 km/hr for the half of the remaining time of the journey, at 12.5 km/hr for the half of the remaining time of the journey and so on until the car accomplishes its journey. The average speed of the car is approximately equal to

(1) 100 km/hr

(2) 50 km/hr

(3) 33 km/hr

(4) 25 km/hr

Question 6

Excluding stoppages, the average speed of a bus is 54 km/hr, and including stoppages it is 45 km/hr. For how many minutes does the bus stop per hour?

(1) 12 min

(2) 10 min

(3) 11 min

(4) 15 min

Question 7

The speeds of a scooter, a car and a train are in the ratio 1 : 4 : 16. If all of them cover equal distance, the ratio of time taken by each of the vehicle is:

(1) 256 : 16 : 1

(2) 1 : 4 : 16

(3) 16 : 4 : 1

(4) 16 : 1 : 4

Question 8

An escalator is moving up at a constant speed. Govind runs at 8 steps/sec and reaches to the bottom in 18 sec. At the same time Sajid runs at 4.25 steps/sec and reaches to the top in 24 sec. Find the speed of the escalator in steps/sec.

(1) 2

(2) 3

(3) 4/19

(4) 1

Question 9

Moody takes 30 seconds to finish riding an escalator if he walks on it at his normal speed in the same direction. He takes 20 seconds to finish riding the escalator if he walks at twice his normal speed in the same direction. If Moody decides to stand still on the escalator, then the time, in seconds, needed to finish riding the escalator is

Question 10

Nikhil reaches school everyday at 5 p.m. to pick up his children. On a Saturday, the school got over at 4 p.m. and the children started walking towards home. Nikhil met them on their way and returned home 30 min earlier. How long did the children walk?

(1) 50 min

(2) 55 min

(3) 45 min

(4) 30 min

Question 11

By walking at 4/5th of his usual speed, a man reaches office 10 minutes later than usual. What is the usual time taken by the man to reach office?

(1) 30 min

(2) 40 min

(3) 35 min

(4) 50 min

Question 12

If a man walks from his house at 5 km/ hr, he reaches office 5 minutes late. If he walks at 6 km/hr, he reaches 5 minutes early. Then the distance from his house to the office is

(1) 3 km

(2) 4km

(3) 5 km

(4) 6km

Question 13

A train travels at a uniform speed of 15 km/hr. If the speed had been 10 km/hr more, it would have taken 8 hours less to cover the distance between stations A and B. What is the distance (in km) between A and B?

(1) 100

(2) 300

(3) 250

(4) 200

Question 14

A train travelled at one-thirds of its usual speed, and hence reached the destination 30 minutes after the scheduled time. On its return journey, the train initially travelled at its usual speed for 5 minutes but then stopped for 4 minutes for an emergency. The percentage by which the train must now increase its usual speed so as to reach the destination at the scheduled time, is nearest to

(1) 58

(2) 67

(3) 50

(4) 61

Question 15

Anu takes 20 min less than the usual time to reach her office if her speed increases by 5 km/hr, and takes 30 min more than the usual time if her speed decreases by 5 km/hr. What is her usual speed?

(1) 20 km/hr

(2) 30 km/hr

(3) 25 km/hr

(4) 35 km/hr

(5) None of these

Question 16

A and B start simultaneously from P and reach R via Q along the same road. A travels at 15 km/hr from P to Q and at 20 km/hr from Q to R. B travels at 20 km/hr from P to Q and at 15 km/hr from Q to R. A reaches at R, 10 min before B reaches there. If the time taken to cover PQ to the time taken to cover QR by B is in the ratio 1:2, find the distance between P and R.

(1) 35 km

(2) 50 km

(3) 55 km

(4) 45 km

Question 17

After travelling for 30 minutes, a train met with an accident and was stopped there for 45 minutes. Due to the accident, its speed reduced to $\frac{2}{3}$rd of its initial speed and the train reached its destination 90 minutes late. Had the accident occurred 60 km after the point it occurred earlier, the train would have reached 30 minutes earlier than it was reaching after the previous accident. Find the length of the journey.

(1) 90 km

(2) 120 km

(3) 60 km

(4) 180 km

Question 18

The distance between Ankit's house and office is 60 km. One day, he covers half of the distance at twice of his usual speed and the rest of the distance at half of his usual speed. If he took 75 minutes more than the usual time, find the usual speed (in km/hr).

(1) 12

(2) 15

(3) 20

(4) 10

Question 19

If a man cycles at 10 km/hr, then he arrives at a certain place at 1 p.m. If he cycles at 15 km/hr from the same starting point, he will arrive at the same destination at 11 a.m. At what speed must he cycle to get there at noon?

(1) 11 km/hr

(2) 12 km/hr

(3) 13 km/hr

(4) 14 km/hr

Question 20

Only a single rail track exists between stations A and B on a railway line. One hour after the north- bound super fast train N leaves station A for station B, a south-bound passenger train S reaches station A from station B. The speed of the super fast train is twice that of a normal express train E, while the speed of a passenger train S is half that of E. On a particular day, N leaves for B from A, 20 min behind the normal schedule. In order to maintain the schedule, both N and S increased their speeds. If the super fast train doubles its speed, what should be the ratio (approximately) of the speeds of passenger train to that of the super fast train so that the passenger train S reaches exactly at the scheduled time at A on that day?

(1) 1 : 3

(2) 1 : 4

(3) 1 : 5

(4) 1 : 6

Thanks!